molecules are scattered more. And waves have trouble ignoring obstacles as big as they are. (You can see this in water waves scattered by the pilings of piers, or bathtub waves from a dripping faucet encountering a rubber duck.) The shorter wavelengths, those that we sense as violet and blue light, are more efficiently scattered than the longer wavelengths—those that we sense as orange and red light. When we look up on a cloudless day and admire the blue sky, we are witnessing the preferential scattering of the short waves in sunlight. This is called Rayleigh scattering, after the English physicist who offered the first coherent explanation for it. Cigarette smoke is blue for just the same reason: The particles that make it up are about as small as the wavelength of blue light.

So why is the sunset red? The red of the sunset is what's left of sunlight after the air scatters the blue away. Since the atmosphere is a thin shell of gravitationally bound gas surrounding the solid Earth, sunlight must pass through a longer slant path of air at sunset (or sunrise) than at noon. Since the violet and blue waves are scattered even more during their now-longer path through the air than when the Sun is overhead, what we see when we look toward the Sun is the residue—the waves of sunlight that are hardly scattered away at all, especially the oranges and reds. A blue sky makes a red sunset. (The noontime Sun seems yellowish partly because it emits slightly more yellow light than other colors, and partly because, even with the Sun overhead, some blue light is scattered out of the sunbeams by the Earth's atmosphere.)

It is sometimes said that scientists are unromantic, that their passion to figure out robs the world of beauty and mystery. But is it not stirring to understand how the world actually works—that white light is made of colors, that color is the way we perceive the wavelengths of light, that transparent air reflects light, that in so doing it discriminates among the waves, and that the sky is blue for the same reason that the sunset is red? It does no harm to the romance of the sunset to know a little bit about it.

Since most simple molecules are about the same size (roughly a hundred millionth of a centimeter), the blue of the Earth's sky doesn't much depend on what the air is made of—as long as the air